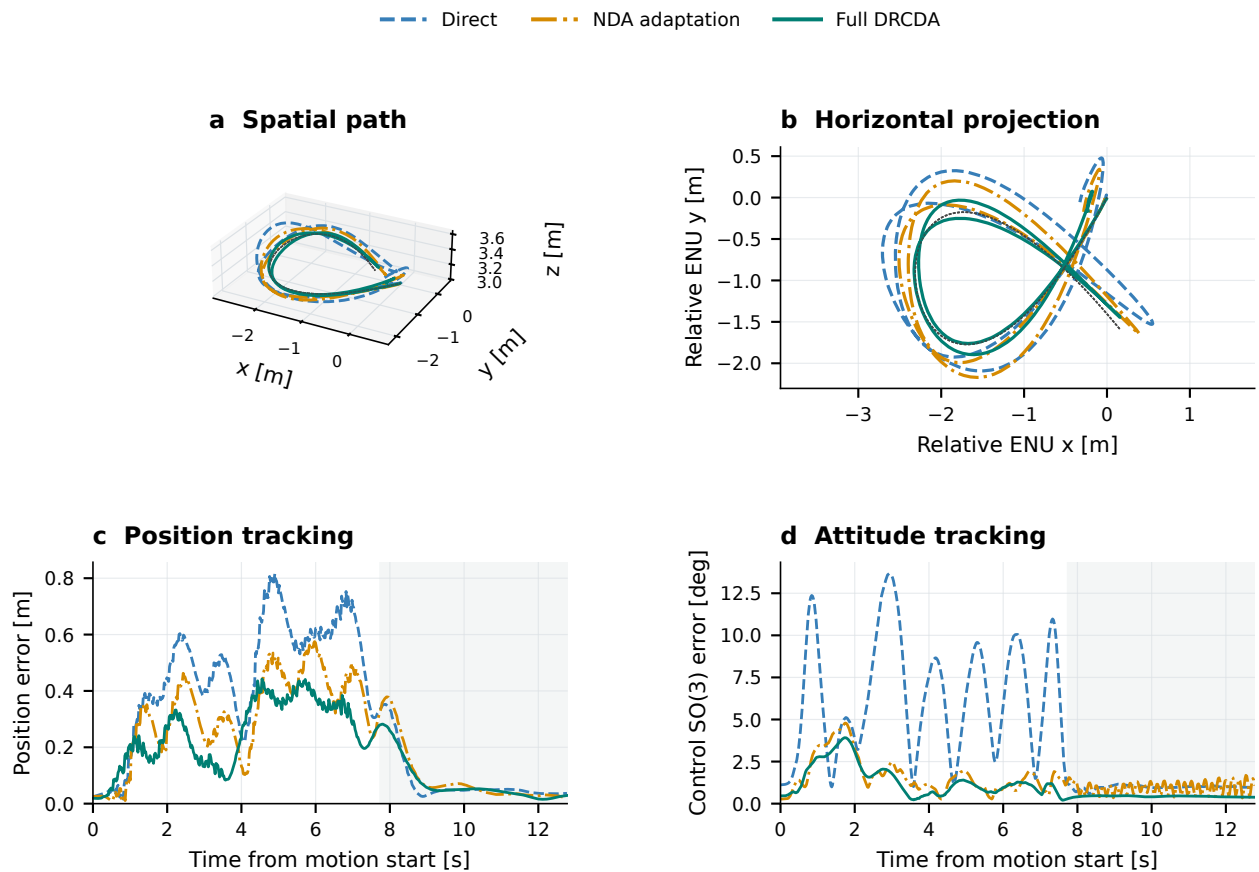
3D Lissajous: matched method comparison

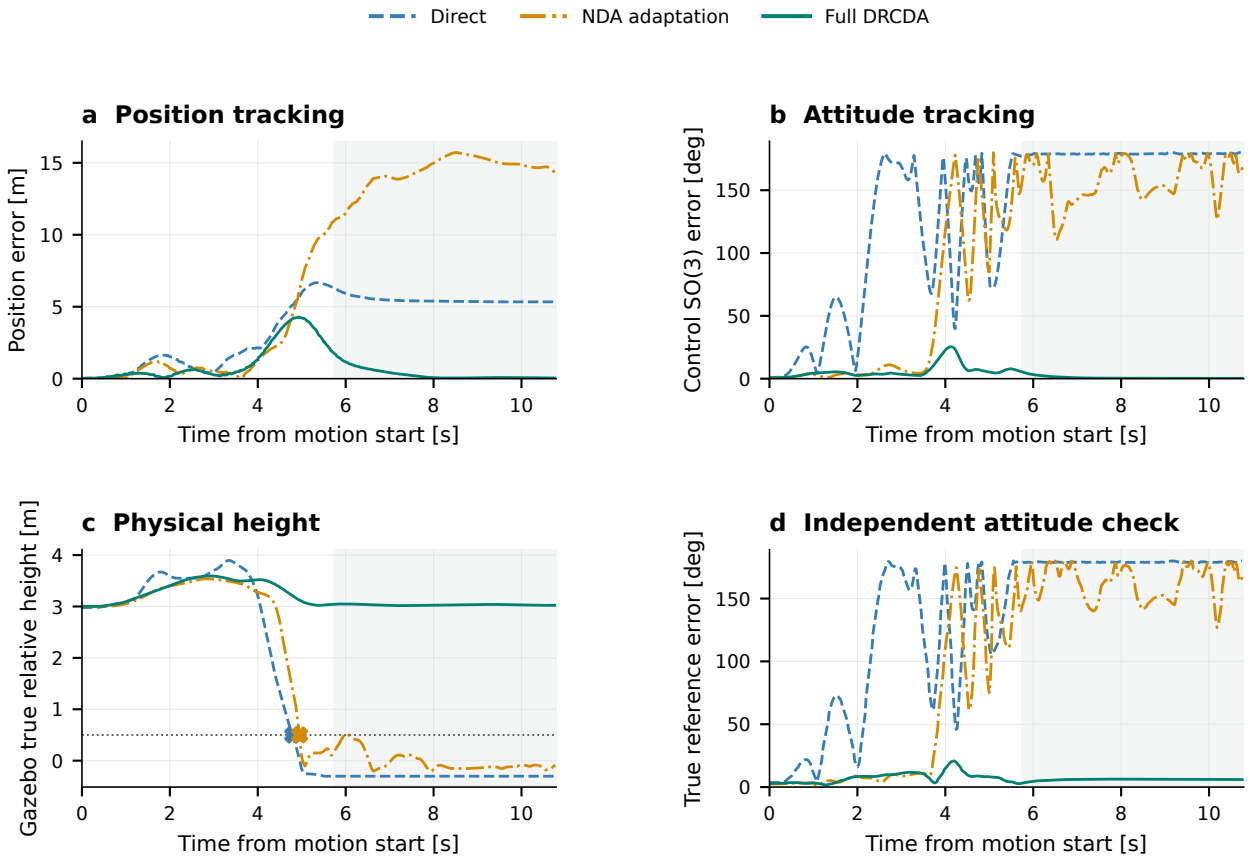
T = 7 s + 0.75 s shared phase ramp | black dotted curves: each run reference



Corrected magnetic frames and unit servo DC gain; no-delay SITL; direct motor/servo commands.
Identical outer-loop gains and limits. One run per condition; no confidence intervals or smoothing.
Gray bands: peak holds (attitude task) or post-maneuver settling (Lissajous).

Aggressive Lissajous: matched comparison

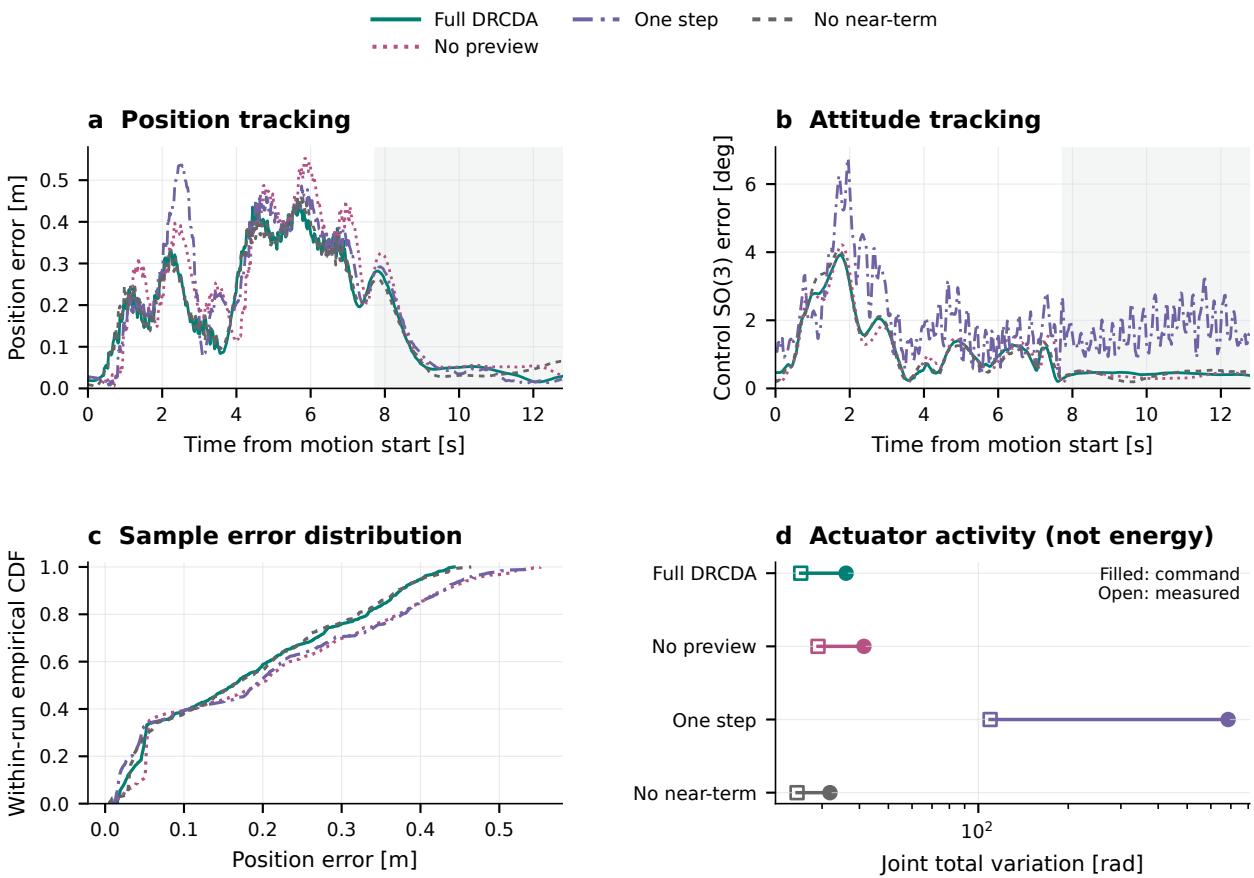
T = 5 s + 0.75 s shared phase ramp | all samples retained, including loss of tracking



Corrected magnetic frames and unit servo DC gain; no-delay SITL; direct motor/servo commands.
Identical outer-loop gains and limits. One run per condition; no confidence intervals or smoothing.
Gray bands: peak holds (attitude task) or post-maneuver settling (Lissajous).

7 s Lissajous: allocation ablations

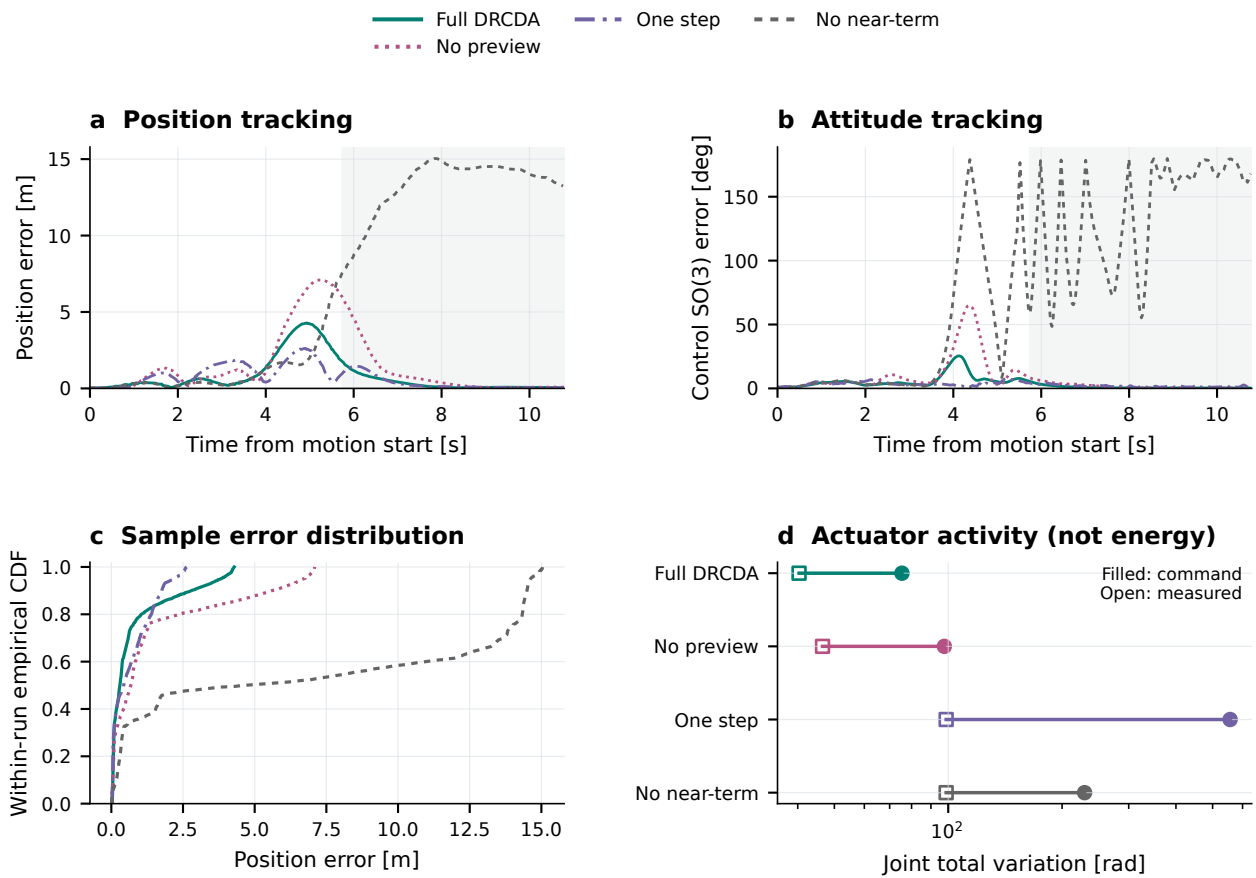
Full / no future reference / one-step horizon / no near-term objective



Corrected magnetic frames and unit servo DC gain; no-delay SITL; direct motor/servo commands.
Identical outer-loop gains and limits. One run per condition; no confidence intervals or smoothing.
Gray bands: peak holds (attitude task) or post-maneuver settling (Lissajous).

5 s Lissajous: allocation ablations

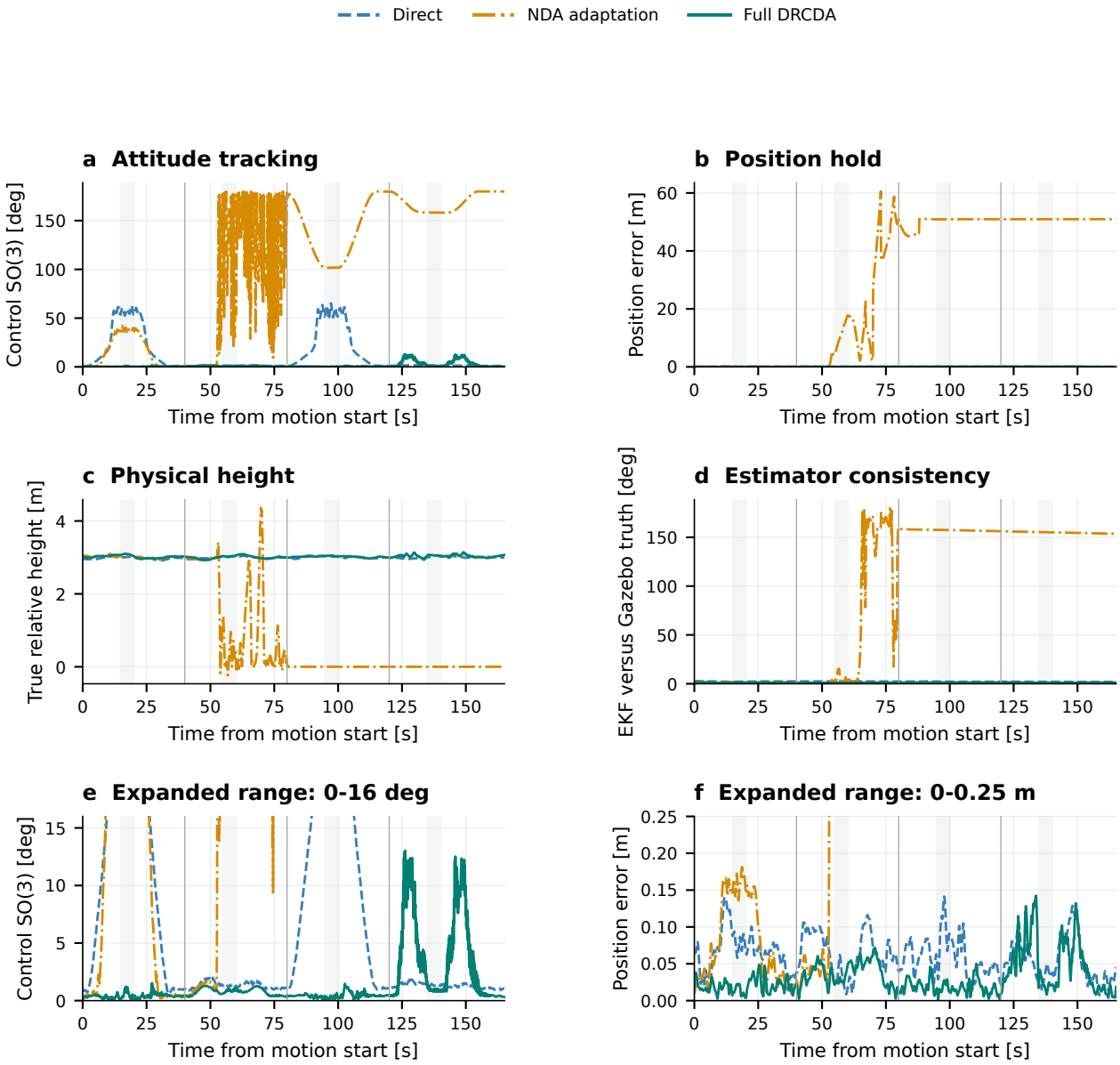
Full / no future reference / one-step horizon / no near-term objective



Corrected magnetic frames and unit servo DC gain; no-delay SITL; direct motor/servo commands.
Identical outer-loop gains and limits. One run per condition; no confidence intervals or smoothing.
Gray bands: peak holds (attitude task) or post-maneuver settling (Lissajous).

Large-attitude method comparison

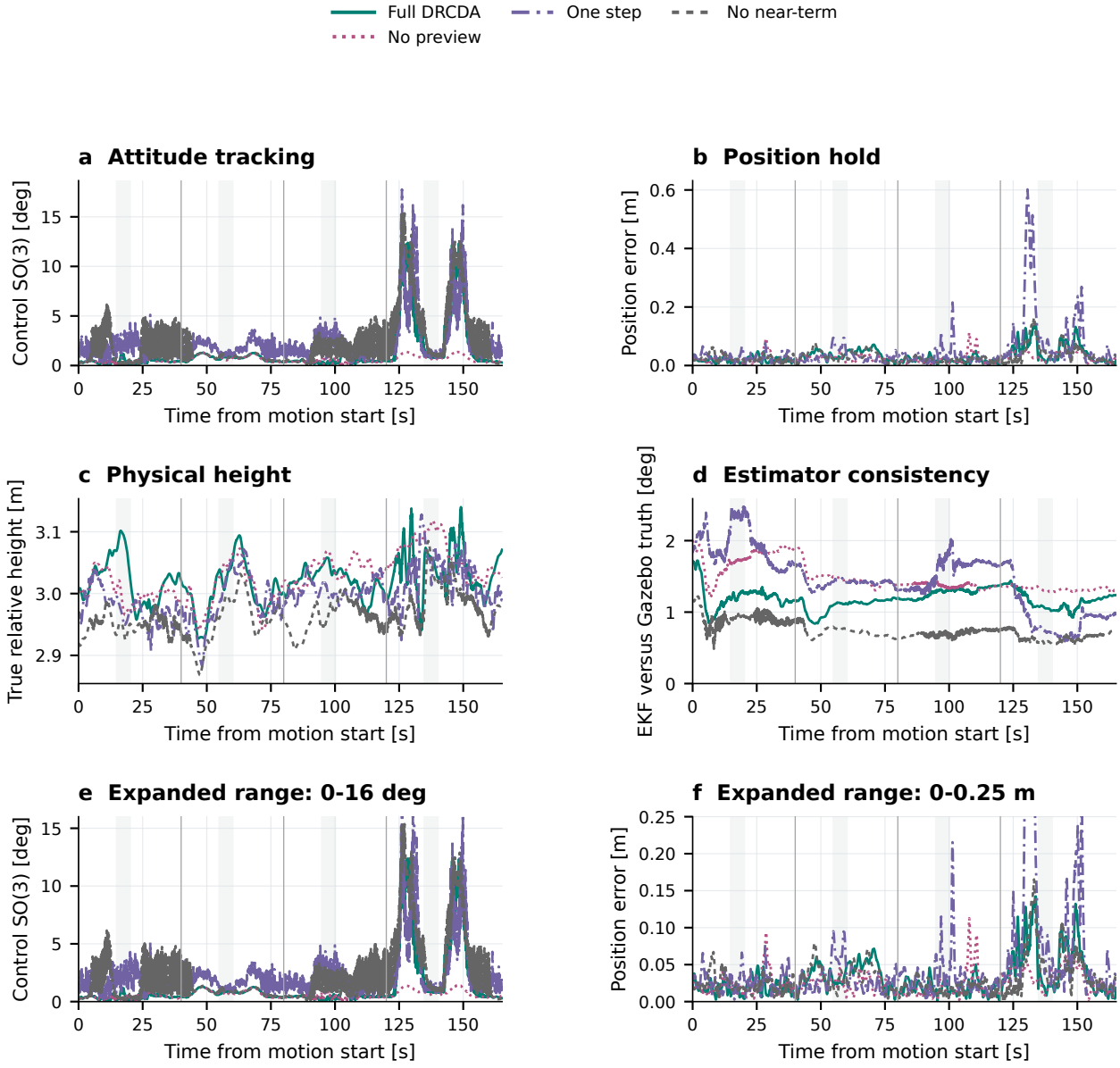
+80 roll / +180 pitch / -80 roll / -180 pitch | 15 s transition, 5 s peak hold



Corrected magnetic frames and unit servo DC gain; no-delay SITL; direct motor/servo commands.
Identical outer-loop gains and limits. One run per condition; no confidence intervals or smoothing.
Gray bands: peak holds (attitude task) or post-maneuver settling (Lissajous).
Panels e-f are labeled range expansions; full failure traces remain in panels a-d.

Large-attitude ablations

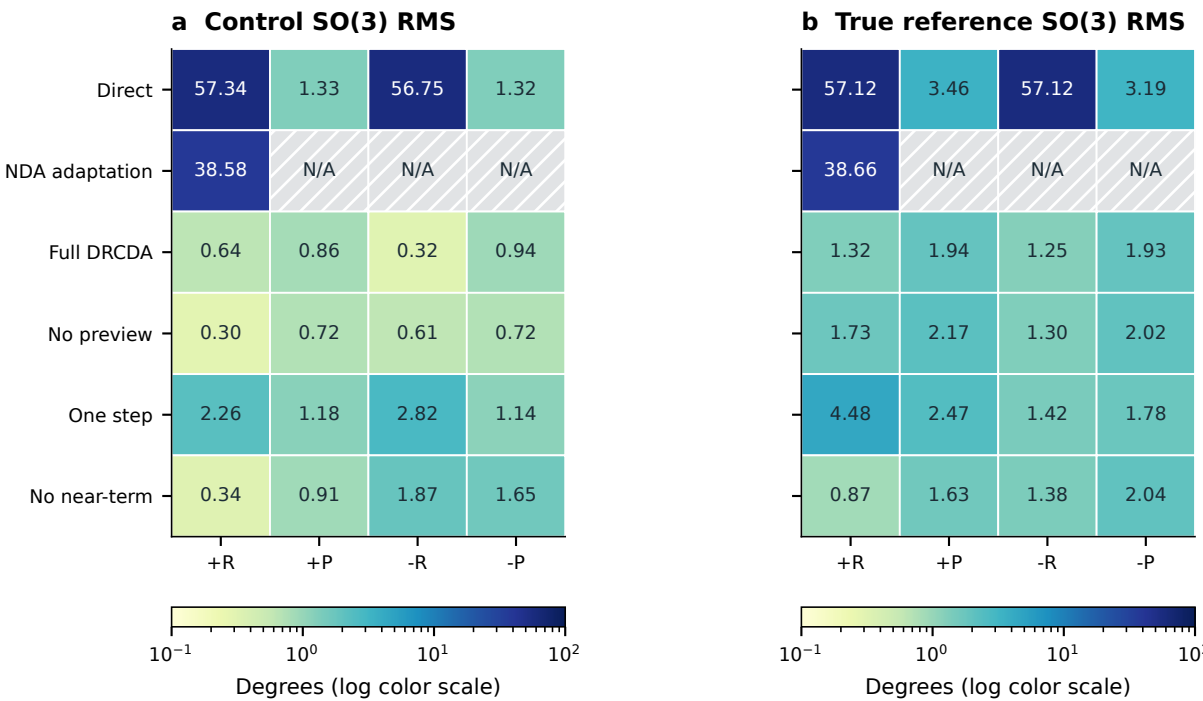
+80 roll / +180 pitch / -80 roll / -180 pitch | 15 s transition, 5 s peak hold



Corrected magnetic frames and unit servo DC gain; no-delay SITL; direct motor/servo commands.
Identical outer-loop gains and limits. One run per condition; no confidence intervals or smoothing.
Gray bands: peak holds (attitude task) or post-maneuver settling (Lissajous).
Panels e-f are labeled range expansions; full failure traces remain in panels a-d.

Large-attitude peak-hold accuracy

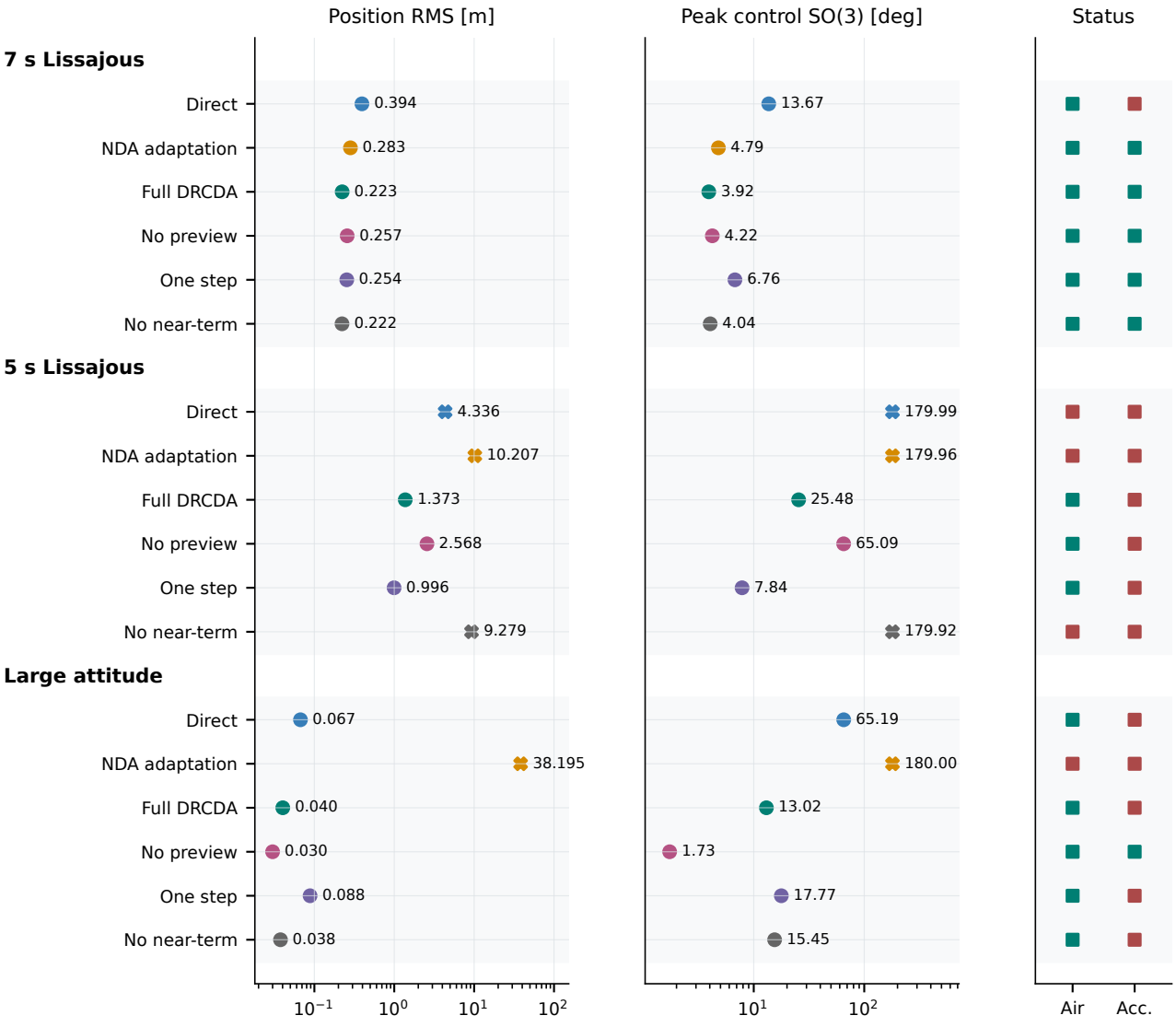
Each cell is the RMS of one 5 s hold; no population statistics



Corrected magnetic frames and unit servo DC gain; no-delay SITL; direct motor/servo commands.
Identical outer-loop gains and limits. One run per condition; no confidence intervals or smoothing.
Hatched cells overlap or follow physical height loss and are not successful hovering evidence.

Matched results at a glance

Single-run results, separated by task | no failure samples removed



Corrected magnetic frames and unit servo DC gain; no-delay SITL; direct motor/servo commands.
Identical outer-loop gains and limits. One run per condition; no confidence intervals or smoothing.
Status: green = yes; red = no. Air = airborne completion; Acc. = original accuracy criteria.
Airborne requires finished/armed, no cutoff, and true height above 0.5 m.